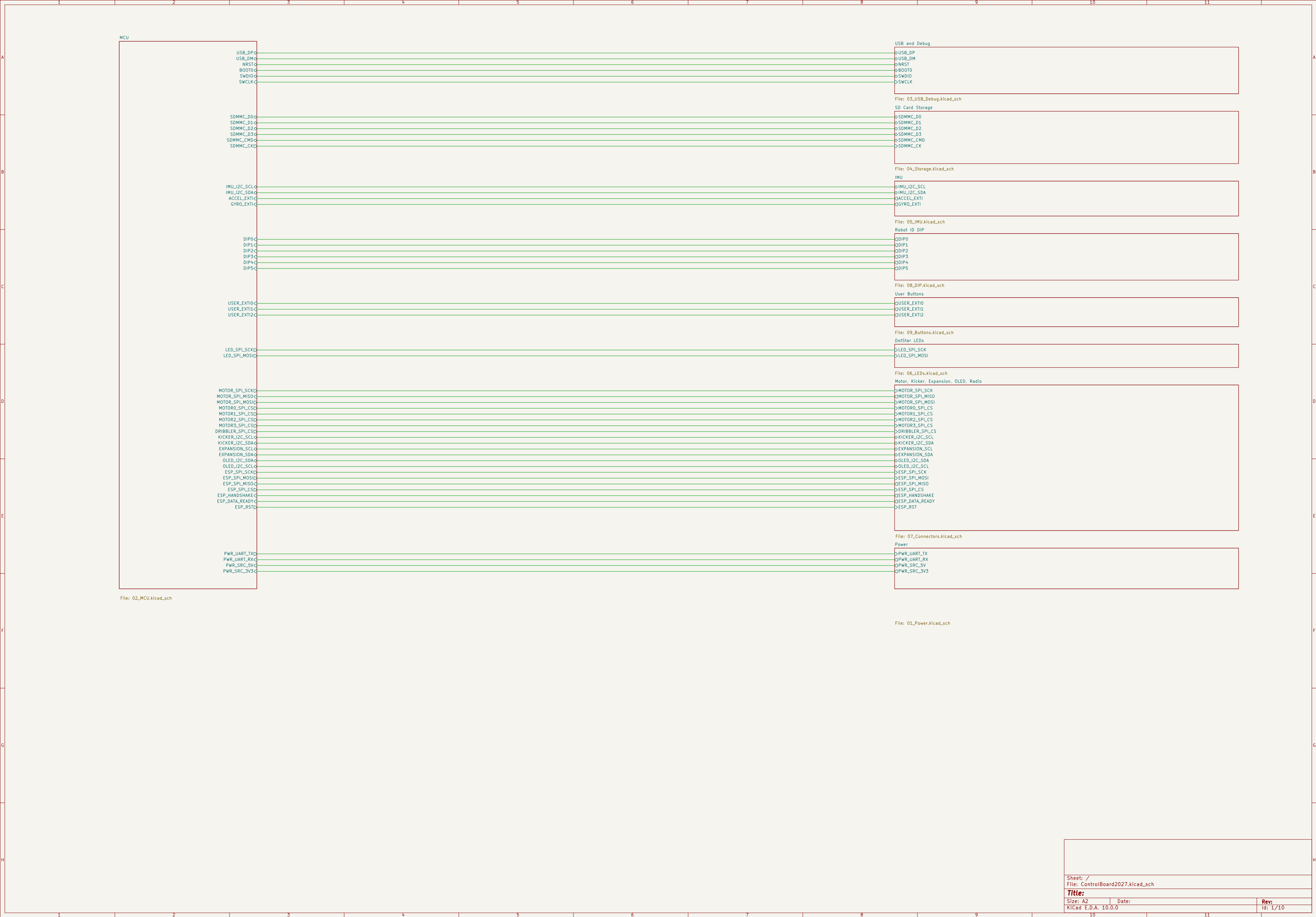
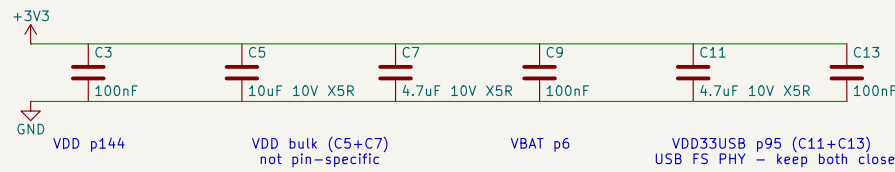
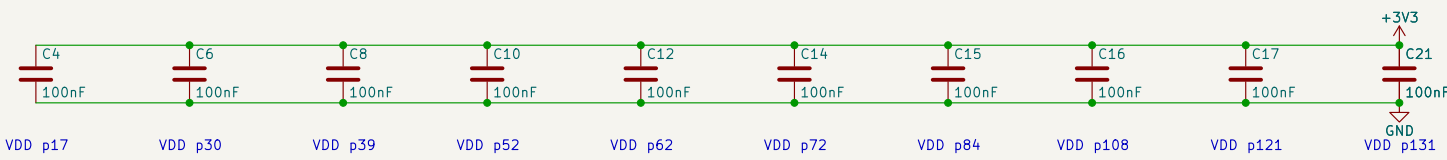




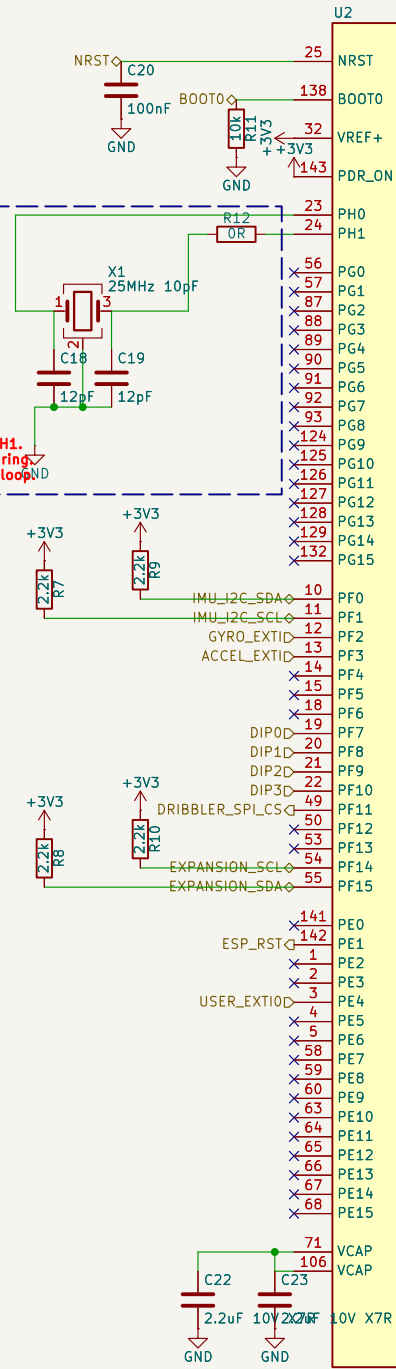
VDD DECOUPLING (EACH CAP AT ITS PIN)

ROUTING – DECOUPLING: each cap beside the STM32 pin named under it, via straight to GND.
C22/C23 (VCAP) tight to p71/p106. The netlist does not keep this pairing.



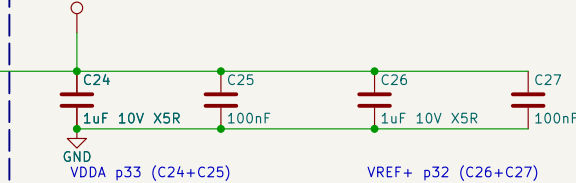
HSE 25MHZ CRYSTAL

ROUTING – HSE CRYSTAL
X1, C18, C19 within 5mm of PH0/PH1.
Short equal OSC traces, GND guard ring.
Nothing routed under or beside the loop.

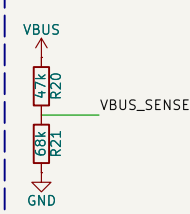


VDDA / VREF+ DECOUPLING

VDDA p33 and VREF+ p32 use +3V3.
Separate local bypass pairs; analog noise must be measured.
Source: DS13313 supply scheme / electrical characteristics.



VBUS SENSE



R15/R16 = 30R; provisional source damping, not certified matching.
No fixed driver resistance or SSR target follows from DS13313 Tables 52/54.
Validate SCK/MOSI/MISO/CS at every cable receiver, both directions.

ROUTING – MOTOR SPI (6 Mbit/s, 5 drops)
R15/R16 within 5mm of PA5/PA7, not at the connectors.

I2C: R7/R9, R8/R10, R14/R17, R18/R19 are each 2.2k.
At 3.3 V: (3.3-0.4)/2.2k = 1.32 mA sink requirement.
For a 300 ns rise-time limit: C <= 300ns/(0.8475R) = 161 pF nominal.
At +5% R: 153 pF. Include ESL pins, cables and parallel pull-ups.
LSM6DSX320X / SSD1306 / DS13313 timing and I/O data; external loads not qualified.

I2C VALIDATION REQUIRED: measure SDA and SCL at worst load.
Do not change to 1.5k from rise time alone: recheck every device's VOL/current.
OLED module pull-ups and expansion-device specifications are not established.

CLOCK: X1 = 25 Mhz; set firmware clock tree to the fitted part.
R12 = 0R is an oscillator drive-tuning link, not line termination.
C18/C19 need measured stray capacitance to establish the 10 pF load.
Startup margin and crystal drive remain unverified (ABM8, DS13313 HSE data).
No blanket claim that 25 Mhz always starts or HS148 is unusable.
SUPPLIES: PDR_ON and VBAT are on +3V3; no backup battery.
VCAP 71/106 share their bypass node; retained pending ERC symbol review.
Do not separate a required supply node merely to silence ERC.
Source: DS13313 power-supply scheme / VCAP data.

USB-C J2: TVS, FUSE, CC RESISTORS, ESD

USB LAYOUT: keep DP/DM coupled over an uninterrupted ground reference.
Join duplicate Type-C data contacts at J2; U3 close to connector.
Final impedance/skew depend on stackup and PHY timing, not series R alone.
DS4260 Figures 6/17: minimize ESD path inductance. Routed 90R pair over In1 GND.

Components and Connections:

- J2 USB-C-DFU**: Connector with pins A4, A5, CC1, CC2, A7, B7, A6, B6, A8, B8, SBU1, SBU2, SH, and A1.
- TVS0500DRV D1**: TVS diode connected between VBUS and GND.
- F1**: Fuse connected between VBUS and the TVS diode.
- MF-MSMF075/16X**: Common Mode Filter connected between VBUS and GND.
- R22**: 3.1kΩ resistor connected between CC1/CC2 and GND.
- R23**: 5.1kΩ resistor connected between A7/B7 and GND.
- USB_DP_CON**: Signal line connected to A6/B6.
- USB_DM_CON**: Signal line connected to A8/B8.
- SHIELD**: Shield connection to SH pin, which is connected to GND.
- A1**: Pin connected to GND.
- PWR_FLAG**: Power flag connection.
- TP3 TP-VBUS**: Test point connected to VBUS.
- VBUS**: Main power line.
- U3 USBLC6-25C6**: ESD protection diode connected between VBUS and GND.
- USB_DM_CON** and **USB_DP_CON**: Signal lines connected to the ESD diode.

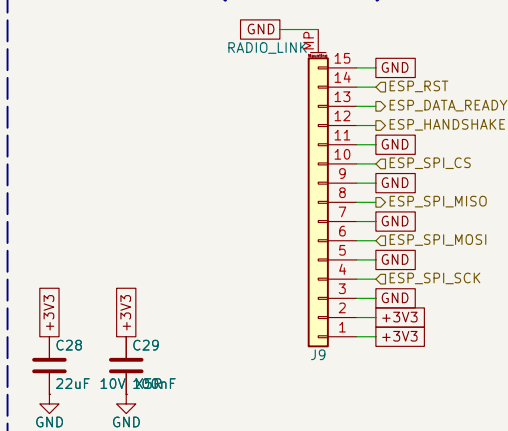
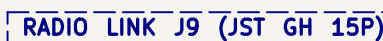
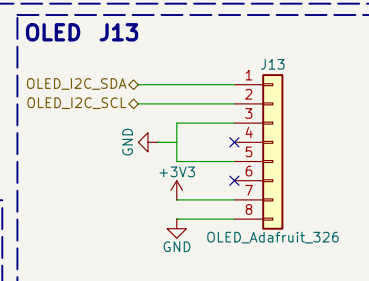
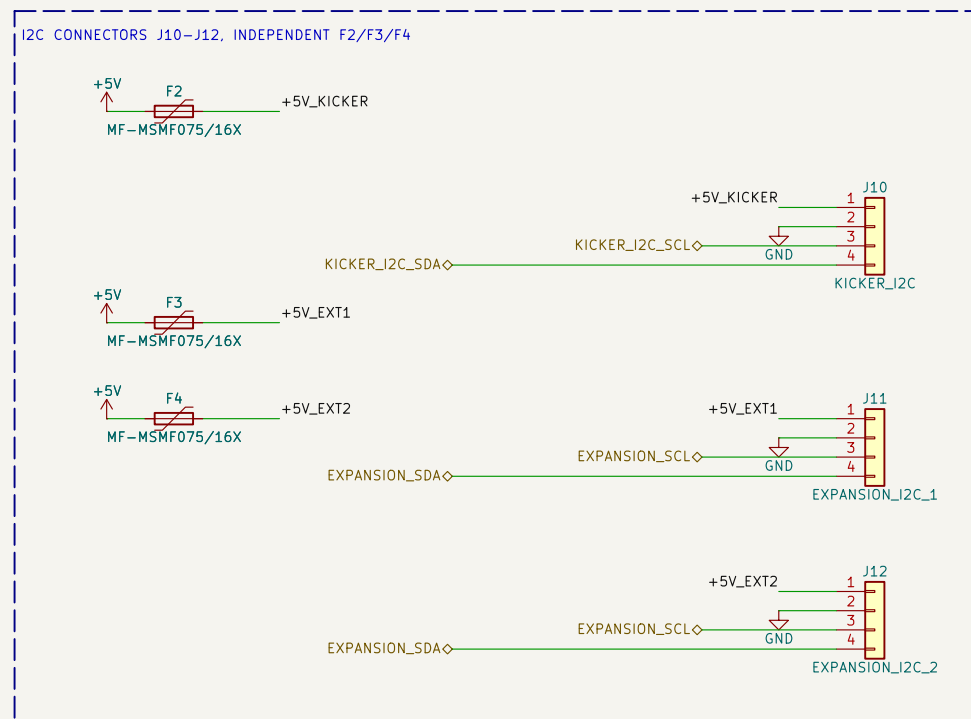
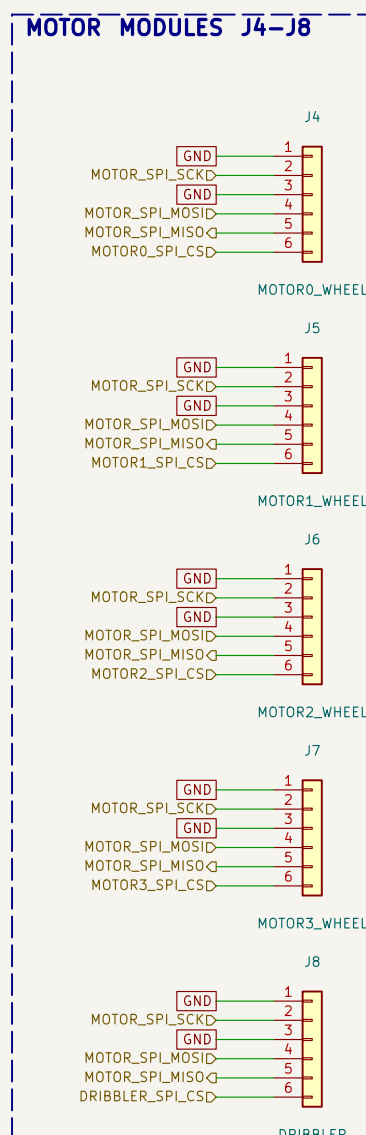
USB LOAD: MF-MSMF075/16X I_{hold} = 0.75 A at 23C, 0.55 A at 50C.
Actual concurrent worst-case board/external current is not established.
Restricted bench use until host current, startup and thermal budget are checked.
F1 does not negotiate USB current; replacing it alone cannot approve more load.

SWD DEBUG J3, RESET AND BOOT BUTTONS

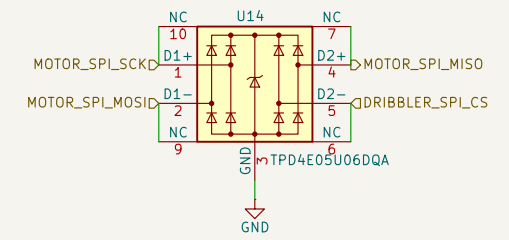
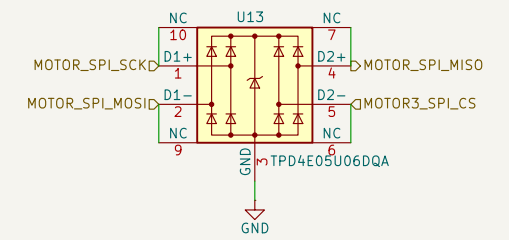
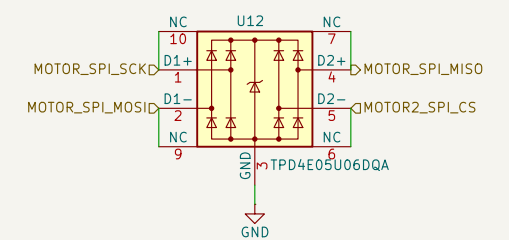
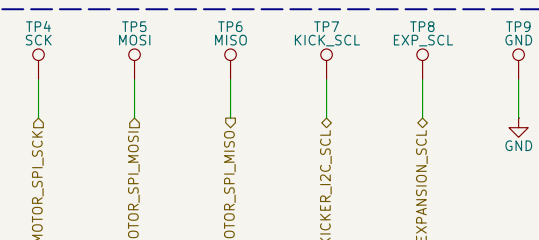
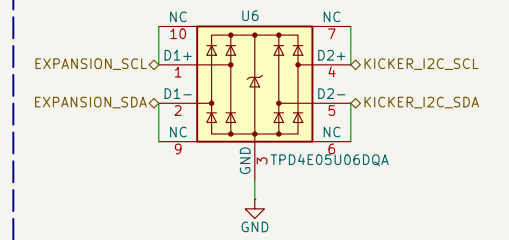
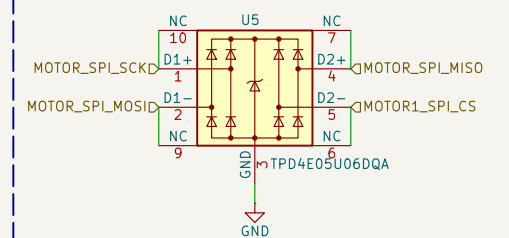
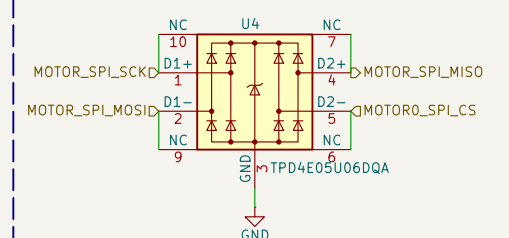
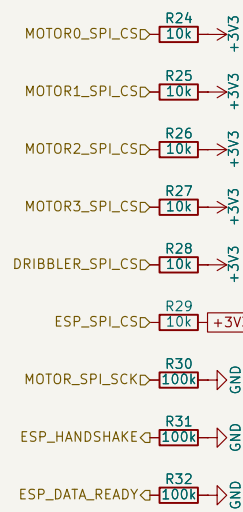
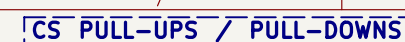
The diagram illustrates the wiring for SWD debug, reset, and boot buttons. A 5-pin connector J3 is used for the SWD interface. Pin 1 is connected to +3V3. Pin 2 is connected to SWCLK. Pin 3 is connected to SWDIO. Pin 4 is connected to GND. Pin 5 is connected to SWD_5pin. SWCLK and SWDIO are connected to a microcontroller. SWDIO is also connected to NRST. NRST is connected to a pull-up resistor SW1 to +3V3 and to a reset button. The reset button is connected to a microcontroller RESET pin and to GND. BOOT0 is connected to a pull-up resistor SW2 to +3V3 and to a boot button. The boot button is connected to a microcontroller BOOT0_DFU pin and to GND.

SWD is direct; damping depends on probe/cable edges and layout.
DS13313 does not prescribe a universal 22R or 55R series resistor.
NRST has an internal 30–50k pull-up; C20 = 100nF (DS13313 NRST data).
U3 clamp rail is +3V3; validate rail injection and power-off behavior.
USBLC6–2 DS4260 pinout/topology; short return path required.

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U3 clamp rail is +V3; validate rail injection and power-off behavior.
USBL6-2 DS4260 pinout/topology; short return path required.



J9 CONTROLBOARD INTERFACE ONLY (remote board not audited):
 1/2 +3V3; 3/5/7/9/11/15/MP GND; 4 SCK; 6 MOSI; 8 MISO;
 10 CS; 14 HARTMATE; 13 DATA READY; 15 RESET
 Configure cable/mating pinout and remote I/O levels before connection.
 Configure reset open-drain only if the remote EN pull-up is present.
 R31/R32 = 100k: bias when disconnected, not guaranteed reset-state logic.
 ESP32-C5 MTDK/GPIO4 has a reset pull-up: receiver level must be checked.
 Protocol success mode and interrupt masking are firmware requirements.

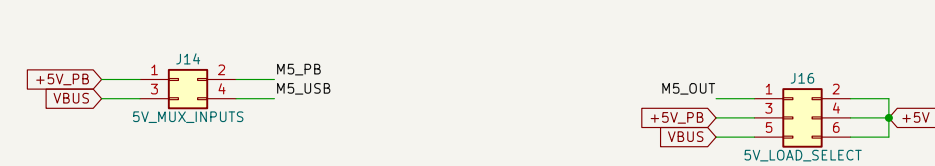


SPI ROUTING: routed -50R over I_{n1} GND: receiver data still pending.
R15/R16 are provisional damping; verify all multidrop endpoints.
Place ESD devices at connectors with short ground returns.
F2 → J10, F3 → J11, F4 → J12: independent PTC output rails.
MF -MSMF075: each holds 0.55 A at 50C: NOT a combined available budget.
Shared U₉/source limits still apply: fault selectivity is not guaranteed.
Shared I_{2C} signals remain vulnerable to any device holding a bus line low.

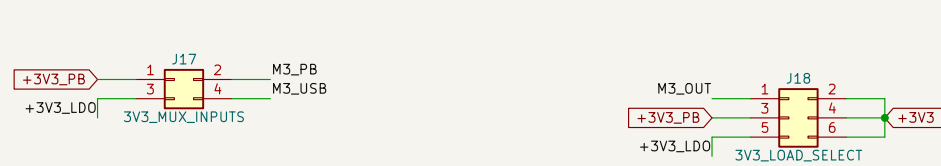
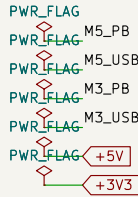


5V POWER PATHS: J14 INPUT LINKS / J16 LOAD SELECT

3V3 POWER PATHS: J17 INPUT LINKS / J18 LOAD SELECT

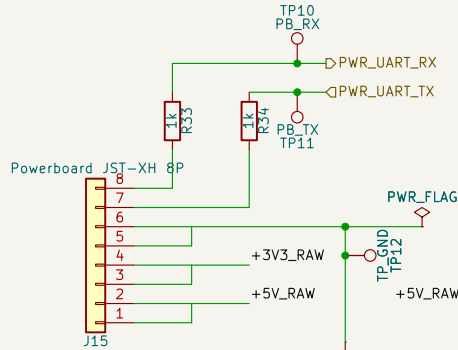


J14/J17: fit BOTH caps for normal operation.
Remove ALL FOUR input caps for failed-mux recovery.

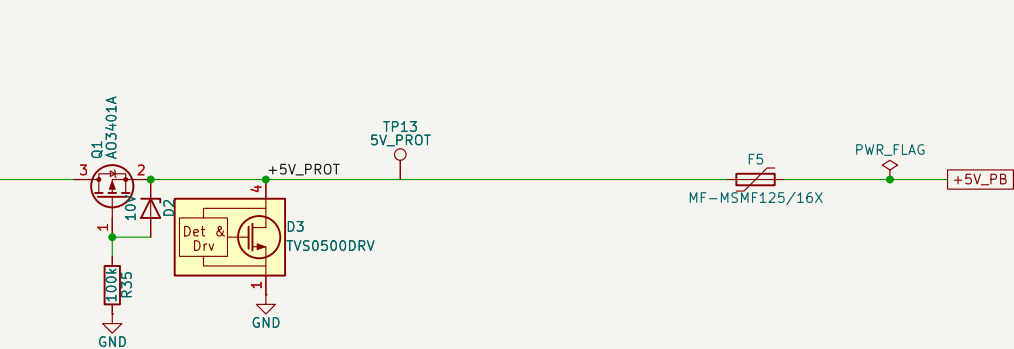


J16/J18: ONE cap each, same position on both.
1-2 MUX | 3-4 POWERBOARD | 5-6 USB. No cap = OFF.
POWER OFF before moving caps. Multiple caps SHORT SOURCES.

POWERBOARD CONNECTOR J15

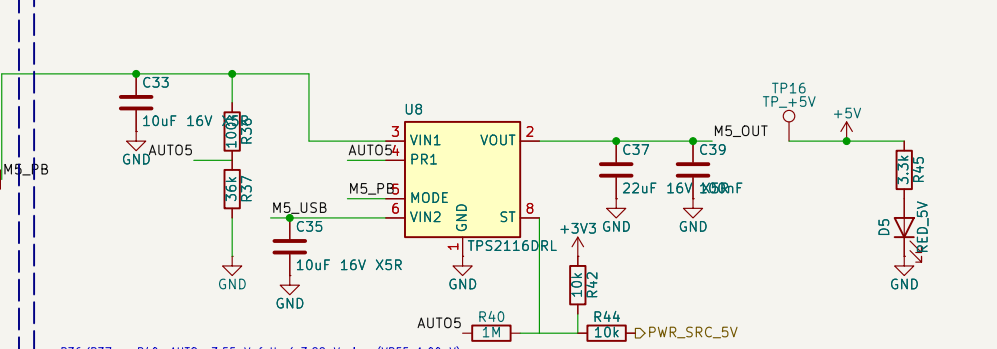


5V INPUT: Q1 REVERSE POLARITY, D3 TVS, F5 FUSE



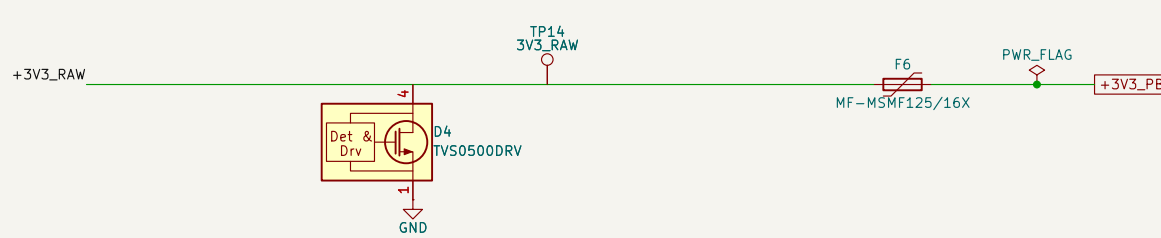
D2: BZX384-C10 gate clamp; characterize transients.
10 V rating is tested at 5 mA, not the R35 bias current.

5V MUX U8: AUTO; BYPASS AT J16

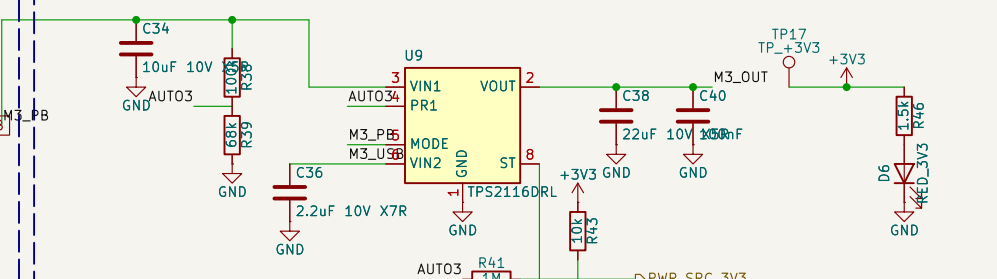


R36/R37 + R40: AUTO -3.55 V fall / 3.88 V rise (VREF 1.00 V)
only with 3.3 V ST bias. VREF tolerance is +/-8%: fall 3.24-3.86 V.

3V3 INPUT: D4 TVS, F6 FUSE

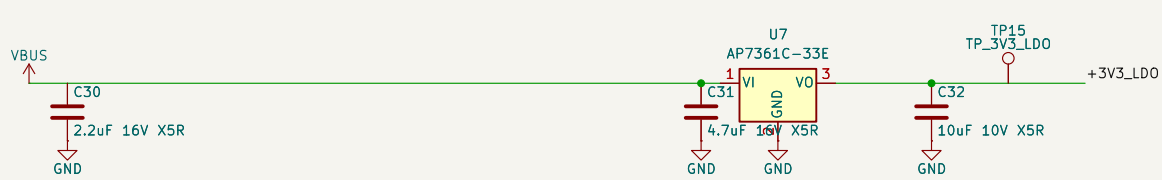


3V3 MUX U9: AUTO; BYPASS AT J18



R38/R39 + R41: ST pull-up is +3V3, which tracks VIN1 while selected.
Nominal fall -2.34 V, rise -2.57 V; NOT rail-valid detection.

USB VBUS AND 3V3 LDO U7



USB INPUT: C30+C31 = 6.9uF directly on VBUS; C35 adds 10uF via J14.
Bypass has no mux soft-start or reverse blocking; validate inrush and load.
PTC rating is not USB-current permission. USB 3V3 still uses U7 LDO.
Sources: TPS2116 7.3.2: MF-MSMF; AP7361C.

PROTECTION LIMITS: TVS0500 clamps up to 9.2 V at 43 A.
This is NOT proof of protection for 6 V-absolute-max TPS2116 or 3V3 loads.
PTCs are thermal devices, not fast current limits; verify fault coordination.
Sources: TVS0500, TPS2116 absolute ratings, Bourns MF-MSMF.

PHYSICAL BYPASS: J16/J18 carry load current, not PR1 control.
For recovery: power OFF, remove all J14/J17 caps, select PB or USB on J16/J18.
Bypass omits mux reverse blocking / soft-start. Source fuses remain.
ST reports MUX state only; cannot identify bypass source or USB current budget.
Control/status resistors remain; NOT galvanic or arbitrary-fault isolation.
Sources: TPS2116 SLVSFG1A 7.3 / 7.6.1; Samtec TSW and SNT datasheets.

J15: 1-2 +5V; 3-4 +3V3; 5-6 GND; 7 TX; 8 RX.
Mate/cable pinout and load remain external-interface requirements.
Q1 polarity agrees with A03401A pinout. No 3V3 reverse-polarity FET.
R33/R34 limit current; they do not guarantee powered-off isolation.
PTC faults and mux source change can still interrupt other loads.

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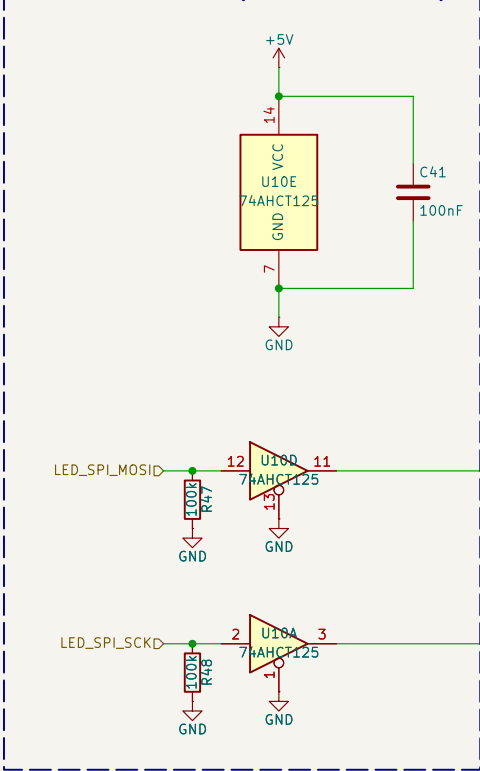
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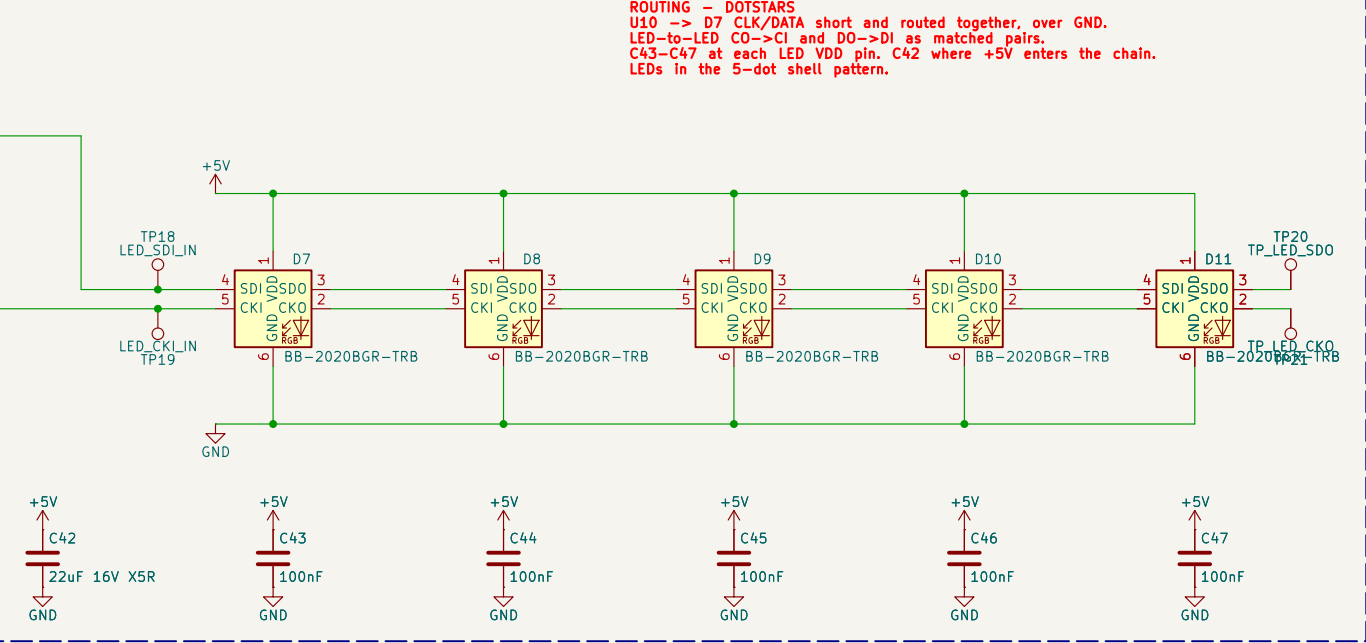
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LEVEL SHIFT U10 (3.3V SPI TO 5V)

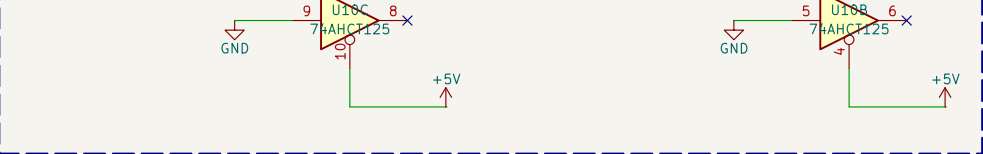


DOTSTAR CHAIN D7-D11 AND DECOUPLING



ROUTING - DOTSTARS
U10 -> D7 CLK/DATA short and routed together, over GND.
LED-to-LED CO->CI and DO->DI as matched pairs.
C43-C47 at each LED VDD pin. C42 where +5V enters the chain.
LEDs in the 5-dot shell pattern.

U10 SPARE GATES



LED LEVEL SHIFT: U10 = SN74AHCT125, specified for VCC 4.5-5.5 V.
3.3 V logic satisfies TTL input levels at a valid U10 supply.
R47/R48 = 100k bias inputs low; unused gates disabled.
D7-D11 = American Bright BB-2020BGR-TRB; use its data sheet/pad drawing.
A muxed USB source does NOT guarantee >=4.5 V after all cable/fuse drops.
Firmware current caps are requirements, not evidence of implemented limits.
Verify LED startup/current, power budget and custom footprints.
TP18/19 check chain input; TP20/21 check chain output.

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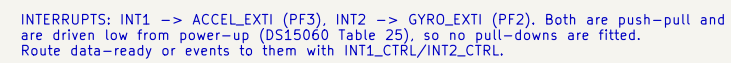
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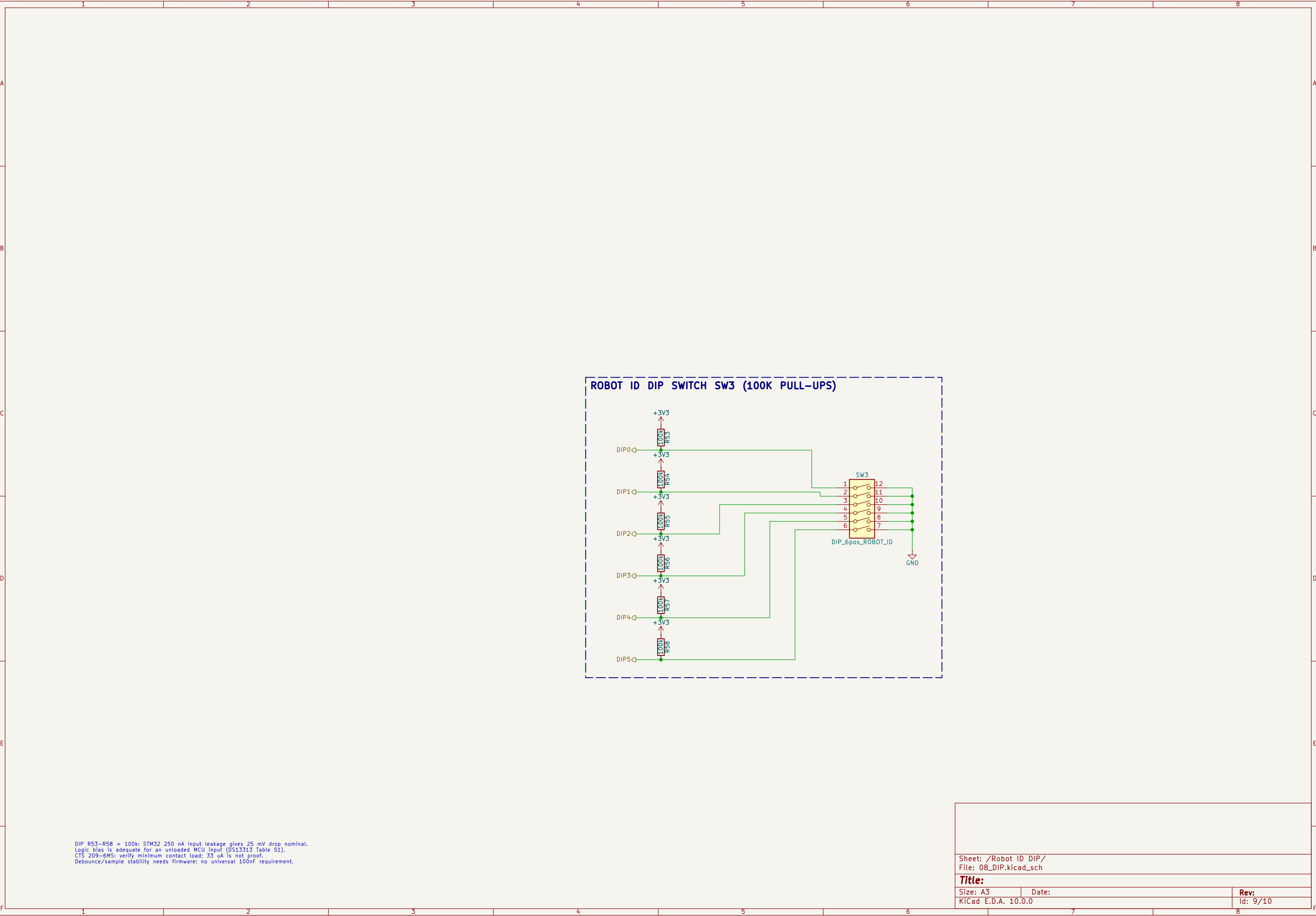
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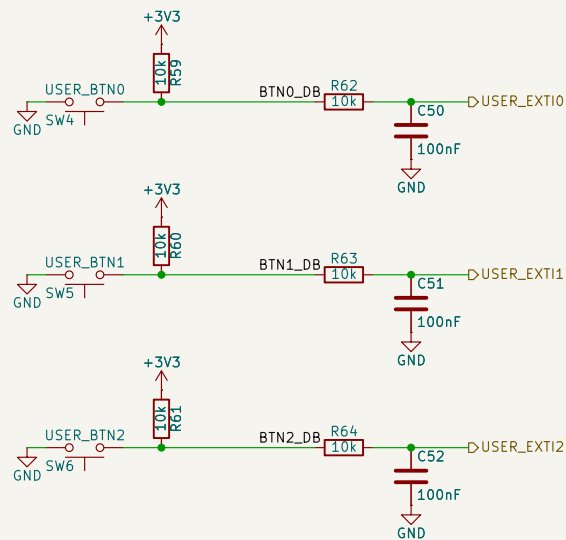
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USER BUTTONS SW4-SW6 WITH RC DEBOUNCE



BUTTONS: R59-R61 pull up with 10k; R62-R64 series 10k; C50-C52 = 100nF.
Calculated nominal time constants: press 1 ms, release 2 ms.
At +/-1% R and +/-10% C: 0.891-1.111 ms and 1.782-2.222 ms.
BOM: R59-R64 are 1%; C50-C52 need a +/-10% X7R part.
RC filtering does not guarantee removal of all bounce; quality switch and firmware.
Series R is between switch node and MCU/cap node, not between cap and MCU.
Input leakage check uses STM32H723 DS13313 Table 51.